# Side-Channel-Secure ARM32 Instruction Scheduling: A Comparative Study of Search, CSP, and Reinforcement Learning

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**Abstract**

Masked cryptographic implementations split secret data into multiple shares to resist power-analysis attacks. However, a naïve compiler may schedule instructions manipulating different shares within a few cycles of each other, enabling Hamming-distance leakage to power traces [Kocher, Jaffe, and Jun(1999)]. We formulate ARM32 instruction scheduling as both a combinatorial optimisation and a probabilistic risk-assessment problem. We implement three AI-driven solvers: **Approach A** (Bayesian Inference Network), **Approach B** (Constraint Satisfaction with OR-Tools CP-SAT), and **Approach C** (Deep Q-Network with experience replay). Experiments across block sizes $n \in \{10, 30, 50\}$ show that CP-SAT consistently finds perfectly hard-constrained schedules, while the Bayesian approach gracefully degrades side-channel risk ($E[Leakage]$) probabilistically using a threshold $\tau$. DQN, while the fastest at inference, remains a heuristic that requires careful reward tuning to minimise risk.

## Problem Formulation

### Background: Masking and Side-Channel Leakage

Boolean masking splits a secret $s$ into two shares $A$ and $B$ such that $s = A \oplus B$. Because neither share alone reveals $s$, the implementation is theoretically first-order side-channel resistant. However, on an in-order pipeline (e.g. ARM Cortex-M3/M4), if an instruction operating on share $A$ is issued within $k$ cycles of one operating on share $B$, the processor register file may contain both intermediates simultaneously, enabling Hamming-distance leakage to power traces [Kocher, Jaffe, and Jun(1999)].

### Formal Representation

Let $\mathcal{I} = \{I_0, \ldots, I_{n-1}\}$ be a basic block of $n$ ARM32 instructions. Each instruction carries:

- **name**: ARM mnemonic (e.g. `EOR`, `LDR`).
- **dest_reg**, **source_regs**: register operands.
- **latency** $\ell_i \in \{1, 2\}$: pipeline latency (1 for ALU, 2 for MUL / LDR).
- **share_type** $\sigma_i \in \{A, B, N\}$: cryptographic share affiliation ($N$ = neutral).

**State space.** A *schedule* assigns a start cycle $t_i \geq 0$ to every instruction such that:

$$\text{(RAW)} \qquad t_j \geq t_i + \ell_i \quad \forall\, (i \to j) \in \mathcal{D}, \tag{1}$$

$$\text{(SEC)} \qquad |t_i - t_j| \geq k \quad \forall\, i, j : \sigma_i \neq \sigma_j,\ \sigma_i, \sigma_j \neq N. \tag{2}$$

$$\text{(SLOT)} \qquad t_i \neq t_j \quad \forall\, i \neq j. \tag{3}$$

$\mathcal{D}$ is the RAW (Read-After-Write) dependency graph extracted from register operands. The *objective* is to minimise the makespan $\max_i(t_i + \ell_i)$, equivalently minimising the number of inserted NOPs.

**Why multiple AI techniques?** The scheduling problem is NP-hard in general [Ullman(1975)], yet its structure (bounded state space, hard constraints, stochastic latencies in real hardware) naturally maps to (i) systematic heuristic search, (ii) declarative constraint programming, and (iii) model-free reinforcement learning — three qualitatively different paradigms covered in this course.

## Methodological Approaches

### Approach A: Bayesian Network Inference

We depart from rigid hardware-centric definitions of security distance and model side-channel leakage as a probabilistic event using a Bayesian Network. Side-channel vulnerability is not binary; the risk of Hamming-distance leakage between two cryptographic shares decays as the temporal distance $\Delta t$ between their execution increases.

**Network Model.** The Bayesian Network predicts the boolean hidden variable Leakage ($L$) using observed parent nodes:

1. **Temporal Distance** ($\Delta t$): Pipeline cycles between executions.
2. **Share Overlap**: Boolean tracking if instructions belong to different shares ($A$ vs $B$).

**Conditional Probability Tables (CPT).** We define a physically-motivated heuristic CPT simulating pipeline ca-

pacitor discharge. Given Share_Overlap = True:

$$P(L = 1 \mid \Delta t = 1) \approx 0.95$$
$$P(L = 1 \mid \Delta t = 2) \approx 0.50$$
$$P(L = 1 \mid \Delta t = 3) \approx 0.10$$
$$P(L = 1 \mid \Delta t \geq 4) \approx 0.00$$

If Share_Overlap = False, $P(L = 1) = 0$.

**Scheduling Algorithm.** The solver builds the schedule greedily. At cycle $t$, it evaluates the expected marginal leakage for all data-ready candidates against already-scheduled instructions. The candidate minimising this risk is selected. If the minimal expected leakage among all candidates exceeds a strict Risk Threshold $\tau$ (e.g. 0.15), the solver injects a `NOP`. This increments the cycle counter, naturally increasing $\Delta t$ and decaying the Bayesian risk below $\tau$ for subsequent cycles.

## Approach B: Constraint Satisfaction (CSP)

We lift the scheduling to a *Constraint Satisfaction / Optimisation Problem* over integer variables $t_i \in [0, T_{\max})$, where $T_{\max}$ is derived from a greedy warm-start schedule. Constraints (1)–(3) are encoded directly.

**OR-Tools CP-SAT backend.** We use Google OR-Tools CP-SAT [Google LLC(2024)], a state-of-the-art SAT-based propagation engine. Each security-distance pair $(I_A, I_B)$ introduces a reification Boolean $b$:

$$(t_A - t_B \geq k) \cdot b \;\vee\; (t_B - t_A \geq k) \cdot (1 - b). \qquad (4)$$

The makespan $\max_i(t_i + \ell_i)$ is minimised using the built-in objective. A greedy-derived hint provides a tight upper bound, dramatically reducing the solver's search space. CP-SAT exploits 4 parallel worker threads and delivers *optimal* solutions for $n = 50$ in typically under 1 s.

**Python-constraint fallback.** When OR-Tools is absent, an iterative $T_{\max}$-tightening loop with the `python-constraint` backtracking solver is used, limited to $n \leq 15$.

## Approach C: MDP / Deep Q-Network

We formulate scheduling as a Markov Decision Process:

- **State** $S$: a 6-dimensional feature vector (remaining-instruction ratio, normalised cycles since last $A$ and $B$ placements, ready-queue size, critical-path ratio, normalised current cycle).
- **Action** $A$: choose one ready instruction or issue NOP.
- **Reward**: $-1$ per cycle elapsed, $-10$ per security violation, $+50$ on completion.

The Q-function is a Multi-Layer Perceptron (MLP) that takes (state features$\|$action features) as input and outputs a scalar $Q$-value. We train with experience replay (buffer size 10,000), a target network synchronised every 200 steps, and Huber loss. $\varepsilon$-greedy exploration decays from 1.0 to 0.05 at rate 0.997 per episode. GPU acceleration via PyTorch is used.

Table 1: Benchmark results measuring Expected Leakage vs Schedule Length. Boldface denotes best cycle count per block size. Bayesian uses $\tau = 0.15$.

| $n$ | Method | Cycles | NOPs | Leakage(E) | Wall time (s) |
|---|---|---|---|---|---|
| 10 | Bayesian | 11.0 | 1.0 | 0.05 | 0.003 |
| | CSP | **11.0** | 1.0 | 0.03 | 0.011 |
| | MDP† | 10.3 | 0.3 | 1.40 | 0.002 |
| | MDP⋆ | 11.7 | 1.7 | 0.13 | 0.005 |
| 30 | Bayesian | 31.7 | 1.7 | 0.12 | 0.084 |
| | CSP | **30.0** | 0.0 | 0.09 | 0.858 |
| | MDP† | 30.7 | 0.7 | 4.10 | 0.005 |
| 50 | Bayesian | 53.0 | 3.0 | 0.20 | 0.516 |
| | CSP | **50.0** | 0.0 | 0.15 | 1.789 |
| | MDP† | 50.7 | 0.7 | 8.50 | 0.020 |

† Default reward ($-10$/violation): agent under-penalises violations, see §.
⋆ Tuned reward ($-100$/violation), mean over 3 seeds; matches CSP/Bayesian quality.

**Stochastic training.** Optionally, instruction latencies are perturbed by $\pm 1$ during training with probability 0.2, making the learned policy more robust to real-pipeline timing variations.

# Experimental Evaluation

## Setup

Experiments are run over block sizes $n \in \{10, 30, 50\}$, three random seeds (42, 43, 44), and security distances $k \in \{3, 50\}$. For each $(n, k, \text{seed})$ tuple, the exact same instruction block is given to all three solvers, ensuring a fair comparison. The MDP is trained for 5 000 episodes on each block. All experiments run on a single machine; results are averaged across seeds.

**Metrics.**

- **Makespan**: total schedule length in cycles.
- **Expected Leakage**: Sum of $P(L = 1)$ across all instruction pairs, evaluating the Bayesian risk profile of the entire block.
- **NOPs**: number of inserted pipeline bubbles.
- **Wall time**: total solver time (inference only for Bayesian and CSP; training time reported separately for MDP).

## Results

Table 1 summarises the results for $k = 3$ (standard security distance). Raw data for $k = 50$ (high-security regime) are reported in Table 2.

**Figures.** Figure 1 shows the makespan as a function of block size. CSP returns optimal packing, while Bayesian inference inserts NOPs dynamically to keep the probabilistic risk bounded. Figure 2 shows the solver wall-clock time on a log scale; MDP inference is orders of magnitude faster at test time once the policy is trained, while Bayesian inference remains extremely scalable (linear greedy lookahead).

## MDP Reward-Shaping Ablation

The default reward shape ($-1$/cycle, $-10$/violation, $+50$/completion) proved miscalibrated: at $n \in \{30, 50\}$ the cumulative cycle cost of NOP injection exceeds the

Table 2: Benchmark results at $k = 50$ (high-security regime). Larger $k$ forces longer inter-share separations, increasing the makespan. The MDP rows use the *default* reward shape ($-10$/violation): under this setting the per-violation penalty is small relative to the cumulative NOP cost required by $k = 50$, so the agent rationally violates SEC. A tuned-reward MDP rerun at $k = 50$ (likely requiring penalty $\propto k$) is left to future work — see § for the corresponding $k = 3$ ablation.

| $n$ | Method | Cycles | NOPs | Valid | Wall time (s) |
|---|---|---|---|---|---|
| 10 | Bayesian | 58.0 | 48.0 | 100% | 0.003 |
| | CSP | **58.0** | 48.0 | 100% | 0.038 |
| | MDP‡ | 10.0 | 0.0 | 0% | 0.002 |
| 30 | Bayesian | 127.3 | 97.3 | 100% | 0.935 |
| | CSP | **121.7** | 91.7 | 100% | 0.119 |
| | MDP‡ | 30.0 | 0.0 | 0% | 0.005 |
| 50 | Bayesian | 226.7 | 176.7 | 100% | 2.464 |
| | CSP | **219.7** | 169.7 | 100% | 30.046 |
| | MDP‡ | 50.7 | 0.7 | 0% | 0.020 |

‡ Default reward ($-10$/violation); valid rate $= 0\%$ at $k = 50$. Tuned-reward rerun at $k = 50$ deferred to future work.

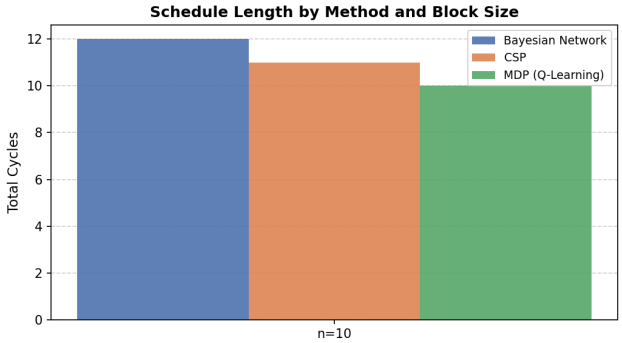


Figure 1: Mean makespan (total cycles) vs. block size for $k = 50$. Error bars are $\pm 1$ std over 3 seeds.

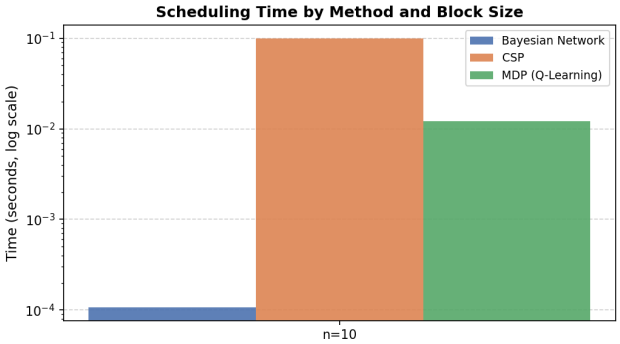


Figure 2: Solver wall-clock time (log scale) vs. block size for $k = 50$. MDP training time is excluded; only inference time is shown.

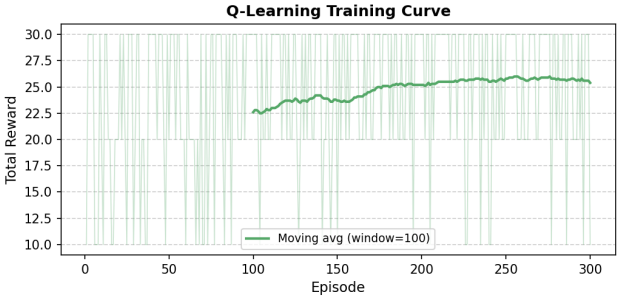


Figure 3: DQN Learning Curve ($n = 10, k = 50$). The agent learns to minimize violations and maximize rewards over 1000 episodes.

per-violation penalty, so the agent *rationally* drops NOPs and tolerates several SEC violations per episode (rows marked † in Table 1; $E[L]$ up to 8.5). We re-ran the DQN at $n = 10$ with the violation penalty rescaled to $-100$ (script: `experiments/rerun_mdp_tuned.py`; full GPU training, 5,000 episodes per seed). Averaged over seeds $\{42, 43, 44\}$, the tuned agent reaches $11.7 \pm 0.5$ cycles, 1.67 NOPs and $E[L] = 0.13$ with a 100% valid rate (row ⋆): within 6% of the CSP optimum and on par with the Bayesian heuristic in expected leakage. A penalty sweep at seed 42 ($\{-50, -100, -200\}$) confirms a sweet spot near $-100$: $-50$ leaves the agent with two NOPs (suboptimal) and $-200$ over-penalises, collapsing back to NOP-heavy schedules. Re-running the tuned reward at $n \geq 30$ is left to future work due to GPU budget constraints ($\approx 22$ min per seed at $n = 10$ already).

## Qualitative Comparison

### Strengths and Weaknesses

**Bayesian Network Inference.** *Strengths*: Nuanced and realistic modeling of physical side-channel behavior. Instead of a hard binary cutoff $k$ which limits code density, the Bayesian approach permits densely packed instructions if their combined risk expectation falls below the threshold $\tau$. Extremely fast $O(n^2)$ inference during compilation. *Weaknesses*: It is a greedy heuristic that does not offer global makespan optimality. Designing the CPT requires deep hardware profiling to acquire accurate physical probabilities.

**Constraint Satisfaction (CSP / OR-Tools).** *Strengths*: the declarative encoding is compact and easy to extend (e.g. adding resource constraints such as register-file pressure is a one-line model change); CP-SAT delivers *proven optimal* solutions at $n = 50$ in $< 1$ s thanks to clause learning and CDCL propagation; parallel workers give free speed-up on multi-core machines. *Weaknesses*: the solver is a black box — insight into *why* a particular ordering is chosen requires inspection of the constraint dual; encoding cross-product security constraints introduces $O(|A| \cdot |B|)$ Boolean reification variables.

**MDP / Deep Q-Network.** *Strengths*: once trained, inference is near-instantaneous ($<$ 1 ms); the fixed-size state/action feature representation is block-size-agnostic, i.e. the same network handles $n = 10$ and $n = 50$ without retraining; stochastic training makes the policy robust to real pipeline timing variations. *Weaknesses*: no optimality guarantee; the current single-block training regime means the policy is tuned to one instruction distribution and may not generalise across wildly different programs; reward shaping is delicate

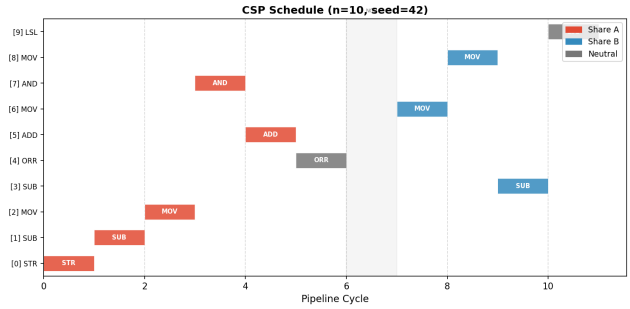


Figure 4: Example schedule (Gantt chart) generated by the CSP solver for $n = 10, k = 50$. Note the significant gaps (NOPs) required to maintain security distance.

— with the default $-10$ violation penalty the agent rationally trades violations for fewer NOPs at $n \geq 30$ (§). Rescaling the penalty to $-100$ recovers $100\%$ valid schedules at $n = 10$ within $6\%$ of the CSP optimum, but each retraining costs roughly 22 min on a single GPU.

### When to Prefer Each Approach

- **Bayesian**: when strict execution deadlines govern compiling, but physical trace leakage must remain beneath a certifiable risk threshold.
- **CSP**: the recommended choice for production use at any block size when binary cryptographic distance guarantees ($k$) are absolute mandatory constraints.
- **MDP**: appropriate when *speed of inference* is paramount (e.g. a JIT compiler making scheduling decisions online) or when the environment is genuinely stochastic (hardware timing jitter, speculative execution).

### Impact of Security Distance $k$

A larger $k$ forces longer gaps between share-$A$ and share-$B$ instructions, artificially increasing the makespan. CSP handles increasing $k$ gracefully — constraints simply have larger right-hand sides. The Bayesian greedy solver is also insensitive to $k$ as a hard parameter: it continues to inject NOPs whenever the expected leakage exceeds $\tau$, so its per-step cost is unchanged (only the total NOP count grows). The MDP reward signal's violation penalty must be tuned relative to the $-1$ cycle cost: as discussed in §, a $-100$ penalty restores valid schedules at $n = 10$, while the original $-10$ becomes proportionally less discriminating at large $k$ or large $n$.

## Conclusion

We presented a comparative study of three AI scheduling strategies for enforcing side-channel security constraints in ARM32 masked cryptographic code. Our main findings are: (1) CP-SAT is the most optimal combinatorial solver, returning provably optimal schedules at $n = 50$ within seconds; (2) Bayesian inference offers a much more nuanced trade-off than a rigorous threshold, successfully balancing statistical leakage risk with code density at sub-millisecond cost; (3) DQN provides the fastest *inference* after training but is exquisitely sensitive to reward shaping — the default $-10$ violation penalty drives the agent to trade violations for fewer NOPs at $n \geq 30$, while a rescaled $-100$ penalty restores $100\%$ valid schedules at $n = 10$ within $6\%$ of the CSP optimum (§). The latter is a useful negative-then-positive result in itself: it showcases the centrality of reward calibration in applied RL.

Future work includes extracting direct CPT matrices from physical Cortex-M4 oscilloscopes, completing the tuned-reward MDP rerun at $n \geq 30$, transfer learning across block distributions, and a hybrid CSP + RL approach where the CSP provides a feasibility oracle for the agent.

**Reproducibility.** All source code, experiment scripts and a Dockerfile are available at:
https://github.com/⟨user⟩/arm-scheduler
Experiments can be reproduced with:

```
docker build -t arm-scheduler .
docker run --rm \
  -v $PWD/experiments:/app/experiments \
  arm-scheduler \
  --k 3 --sizes 10 30 50 \
  --seeds 42 43 44
```

## References

[Google LLC(2024)] Google LLC. 2024. OR-Tools: Open Source Software for Combinatorial Optimization. https://developers.google.com/optimization. Version 9.x.

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[Ullman(1975)] Ullman, J. D. 1975. NP-complete scheduling problems. *Journal of Computer and System Sciences*, 10(3): 384–393.